

AWS Services Guide

A comprehensive guide to the **50+ most important AWS services** you need to know to be an effective AWS practitioner.

Table of Contents

- [Introduction](#)
 - [Core Web Hosting](#)
 - [Traffic Management & Security](#)
 - [Authentication](#)
 - [Compute Services](#)
 - [Storage & Databases](#)
 - [Artificial Intelligence & Machine Learning](#)
 - [Application Coordination & Messaging](#)
 - [Data Analytics](#)
 - [Monitoring & Management](#)
 - [CI/CD & Developer Tools](#)
 - [Identity & Networking](#)
 - [Quick Reference](#)
-

Introduction

AWS offers **over 300 different services**, but the truth is: **nobody knows all of them**. In fact, there are **50 or less services** that you need to know about, with even fewer that you need to know well, to be an effective AWS practitioner.

This guide walks through the essential AWS services using a **practical e-commerce application** (like Amazon.com) as an example, covering all the components you need for a successful internet application.

Core Web Hosting

Everything with an internet-based application starts with a web browser navigating to a domain. Here are the foundational services for hosting web applications.



Route 53

DNS, Domain Management & Traffic Management

- Register and manage domain names
- Set up health checks for your infrastructure
- Geographic routing, latency-based routing, and failover support
- Points users to the right endpoint based on location and availability

Use Case: Register `example.com` and route traffic to the nearest server based on user location.



Amazon S3 (Simple Storage Service)

General Blob Storage Service

- Store static content: images, HTML, text files, videos, backups
- Extremely flexible and inexpensive
- Virtually unlimited scalability
- Can serve static websites directly

Use Case: Host all your product images, CSS, JavaScript, and static HTML files for your e-commerce site.



CloudFront

Content Delivery Network (CDN)

- Caches and replicates content to edge locations worldwide
- Routes traffic to the nearest location for reduced latency
- Integrates seamlessly with S3
- Supports custom SSL certificates

Use Case: Deliver your product images and videos to users globally with low latency, regardless of their location.

Traffic Management & Security

Once users hit your endpoints, you need intelligent traffic routing and protection.



Elastic Load Balancer (ELB)

Application & Network Load Balancing

- Horizontally scale your application
- Distribute incoming API calls to multiple backend instances
- Auto-scaling group integration
- Sticky session support
- Types: Application Load Balancer (ALB), Network Load Balancer (NLB), Classic Load Balancer

Use Case: Distribute traffic across 10 EC2 instances running your web application for high availability.



API Gateway

The "Front Door" of AWS

- Create RESTful and WebSocket APIs
- Forward requests to Lambda, EC2, or any AWS service
- Rate limiting and throttling
- User authentication and authorization
- Request/response transformation
- API versioning

Use Case: Create a REST API with endpoints like `/products` , `/orders` , `/users` that trigger Lambda functions.



AWS WAF (Web Application Firewall)

Application Layer Protection

- Protect APIs from bot attacks
- SQL injection and XSS protection
- IP blocking and geo-blocking
- Custom rule creation
- Integration with CloudFront and ALB

Use Case: Block malicious bots trying to scrape your product catalog or perform credential stuffing attacks.

AWS Shield

DDoS Protection

- **Shield Standard:** Free, automatic protection against common DDoS attacks
- **Shield Advanced:** 24/7 DDoS response team, cost protection, advanced detection
- Protects against Layer 3, 4, and 7 attacks

Use Case: Protect your mission-critical e-commerce application from massive volumetric DDoS attacks.



AWS Certificate Manager (ACM)

SSL/TLS Certificate Management

- Provision, manage, and deploy SSL/TLS certificates
- Free certificates for AWS services
- Automatic renewal
- Integration with Route 53, CloudFront, ELB

Use Case: Enable HTTPS on your domain with a free, auto-renewing SSL certificate.

Authentication

Secure your application by managing who can access it and what they can do.



Amazon Cognito

User Identity & Access Management

- **User Pools:** User registration, login, and directory
- **Identity Pools:** Federated access to AWS resources
- OAuth 2.0 and SAML support
- Multi-factor authentication (MFA)
- Social and enterprise identity providers (Google, Facebook, SAML)

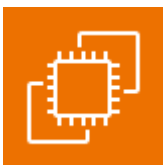
Authentication vs Authorization:

- **Authentication:** Who are you? (Login credentials)
- **Authorization:** What can you do? (Permissions/tokens)

Use Case: Allow users to sign up, log in, and access their order history securely.

Compute Services

Run your application code on various compute platforms based on your needs.



Amazon EC2 (Elastic Compute Cloud)

Virtual Servers in the Cloud

- Rent virtual instances (Linux, Windows, macOS)
- Complete control over OS and applications
- Wide variety of instance types (compute, memory, GPU optimized)
- Manual management of patching, scaling, and availability

Use Case: Run your Node.js web application on an Ubuntu server.



Amazon Lightsail

Simplified Virtual Private Servers

- "EC2 for beginners"
- Predefined application templates (WordPress, LAMP, Node.js)
- Fixed monthly pricing
- Easy-to-use management console

Use Case: Launch a WordPress blog in under 5 minutes with a predictable monthly cost.

Amazon ECS (Elastic Container Service)

Docker Container Orchestration

- Deploy and manage Docker containers
- Integration with ECR (Elastic Container Registry)
- Task definitions and service management
- Supports EC2 and Fargate launch types

Use Case: Run your microservices architecture with isolated Docker containers.



AWS Fargate

Serverless Container Compute

- Run containers without managing servers
- Pay only for resources used by containers
- No EC2 instance management
- Works with both ECS and EKS

Use Case: Deploy containerized applications without worrying about the underlying infrastructure.

Amazon EKS (Elastic Kubernetes Service)

Managed Kubernetes

- Run Kubernetes on AWS without managing control plane
- Automatic upgrades and patching
- AWS service integrations
- Multi-AZ for high availability

Use Case: Deploy and scale your Kubernetes workloads using familiar kubectl commands.



AWS Lambda

Serverless Function Compute

- Deploy code without managing servers
- Automatic scaling (0 to thousands of concurrent executions)
- Pay only for compute time used
- Event-driven architecture
- Supports multiple languages (Node.js, Python, Java, Go, .NET, Ruby)

Use Case: Process uploaded images automatically when users add product photos to S3.

Storage & Databases

Store and manage your application data efficiently.

Storage Services



Amazon EFS (Elastic File System)

Shared File Storage

- NFS-compatible shared file system
- Mount to multiple EC2 instances simultaneously
- Automatic scaling
- Regional availability

Use Case: Share uploaded files across multiple web servers.



Amazon EBS (Elastic Block Store)

Block Storage Volumes

- Acts like a hard drive for EC2 instances
- High-performance for transactional databases
- Attached to a single instance
- Snapshots for backups
- SSD and HDD options

Use Case: Attach high-performance storage to your database EC2 instance.



AWS Secrets Manager

Secure Credential Storage

- Store API keys, database passwords, tokens
- Encrypted at rest and in transit
- Automatic rotation for supported databases
- Fine-grained access control via IAM

Use Case: Store your Stripe API key securely and retrieve it programmatically in your Lambda function.

Relational Databases



Amazon RDS (Relational Database Service)

Managed SQL Databases

- Supports MySQL, PostgreSQL, MariaDB, Oracle, SQL Server
- Automated backups and point-in-time recovery
- Multi-AZ deployments for high availability
- Read replicas for scalability
- Automatic patching

Use Case: Run a PostgreSQL database for your product catalog without managing database administration.



Amazon Aurora

AWS-Built High-Performance Database

- MySQL and PostgreSQL compatible
- 5x faster than MySQL, 3x faster than PostgreSQL
- **Aurora Serverless:** Auto-scales capacity based on demand
- Global databases for multi-region applications
- Automated failover in under 30 seconds

Use Case: Handle unpredictable traffic spikes with Aurora Serverless that scales from 0 to hundreds of thousands of transactions per second.

NoSQL & Specialized Databases



Amazon DynamoDB

Serverless NoSQL Key-Value Store

- Single-digit millisecond latency
- Fully managed, auto-scaling
- Global tables for multi-region replication
- On-demand or provisioned capacity
- DynamoDB Streams for change data capture

Use Case: Store user session data for instant retrieval across millions of users.



Amazon DocumentDB

Managed MongoDB-Compatible Database

- Fully managed document database
- MongoDB compatibility
- Automatic scaling and backups
- Separated compute and storage

Use Case: Migrate your MongoDB workloads to a fully managed service.



Amazon Keyspaces

Managed Apache Cassandra

- Cassandra-compatible API
- Serverless, auto-scaling
- Single-digit millisecond latency
- No server management

Use Case: Run time-series IoT data workloads at massive scale.



Amazon Neptune

Graph Database

- Purpose-built for highly connected data
- Supports Gremlin and SPARQL query languages
- Fast relationship queries
- High availability with read replicas

Use Case: Build a social network feature showing "friends of friends" or a recommendation engine.



Amazon OpenSearch

Search & Log Analytics

- Formerly Amazon Elasticsearch Service
- Full-text search, log aggregation, analysis
- Fuzzy search capabilities (autocomplete)
- Kibana for visualization
- Security, alerting, and SQL support

Use Case: Implement product search with autocomplete and typo tolerance, or aggregate application logs.



AWS DMS (Database Migration Service)

Database Migration

- Migrate databases to AWS
- Minimal downtime migrations
- Supports homogeneous (Oracle to Oracle) and heterogeneous (Oracle to PostgreSQL) migrations
- Continuous data replication

Use Case: Migrate your on-premise Oracle database to Aurora with near-zero downtime.

Caching



ElastiCache

Managed In-Memory Cache

- Supports **Redis** and **Memcached**
- Microsecond latency
- Session storage, leaderboards, caching
- ⚠️ **Note:** If nodes fail, cache is lost (not durable)

Use Case: Cache frequently accessed product information to reduce database load.



Amazon MemoryDB for Redis

Durable In-Memory Database

- Redis-compatible
- **Durable:** Uses transaction log for data persistence
- Microsecond read, single-digit millisecond write latency
- Multi-AZ with automatic failover
- Data survives node restarts

Use Case: Store shopping cart data with Redis-like speed but with durability guarantees.

Artificial Intelligence & Machine Learning

Leverage AI/ML capabilities without deep expertise.



Amazon Bedrock

Foundational Models as a Service

- Access to LLMs: Claude, Llama, Titan, Jurassic
- API-based interaction
- Build chatbots, content generation, summarization
- Fine-tuning capabilities

Use Case: Build a customer service chatbot powered by Claude.

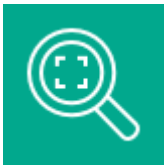


Amazon SageMaker

Complete ML Platform

- Build, train, and deploy ML models
- Jupyter notebooks for data exploration
- Built-in algorithms and frameworks
- Model monitoring and debugging
- MLOps capabilities

Use Case: Train a recommendation model on customer purchase history and deploy it as an API.

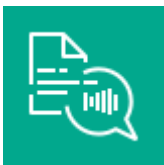


Amazon Rekognition

Image & Video Analysis

- Face detection, comparison, and search
- Object and scene detection
- Text extraction from images (OCR)
- Content moderation
- Celebrity recognition

Use Case: Automatically detect inappropriate product images uploaded by sellers.



Amazon Polly

Text-to-Speech

- Convert text to lifelike speech
- Multiple languages and voices
- Neural TTS for natural-sounding speech
- SSML support for pronunciation control

Use Case: Create audio descriptions of products for accessibility.



Amazon Transcribe

Speech-to-Text

- Automatic speech recognition (ASR)
- Real-time and batch transcription
- Speaker identification
- Custom vocabularies
- Subtitle generation

Use Case: Automatically generate subtitles for product demo videos.

Application Coordination & Messaging

Decouple and coordinate distributed application components.



Amazon SNS (Simple Notification Service)

Pub/Sub Messaging

- Publisher sends message to a topic
- All subscribers receive the message
- Subscribers: Email, SMS, Lambda, SQS, HTTP endpoints
- Fan-out pattern for broadcasting

Use Case: When an order is placed, notify email service, SMS service, and inventory service simultaneously.



Amazon SQS (Simple Queue Service)

Message Queue

- Decouple application components
- Messages wait in queue until consumed
- Standard (at-least-once, best-effort ordering) or FIFO (exactly-once, strict ordering)
- Dead letter queues for failed messages

Use Case: Queue order processing tasks so that sudden traffic spikes don't overwhelm your backend.



Amazon EventBridge

Event Bus & Event Routing

- Route events between AWS services, SaaS apps, and custom applications
- Event rules and patterns
- Schedule expressions (cron jobs)
- Event replay and archive

Use Case: Trigger a Lambda function every day at midnight to generate sales reports.



EventBridge Scheduler

Time-Based Task Scheduling

- Schedule one-time or recurring tasks
- Flexible scheduling (e.g., "3 hours from now")
- Integration with all AWS services
- More advanced than CloudWatch Events

Use Case: Send a review request email 3 days after order delivery.

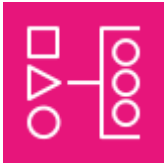


AWS Step Functions

Visual Workflow Orchestration

- Build state machines to coordinate distributed apps
- Visual workflow designer
- Parallel execution, error handling, retries
- Integration with Lambda, ECS, Batch, DynamoDB, SNS, SQS

Use Case: Orchestrate a multi-step order fulfillment workflow: validate payment → update inventory → send confirmation email → schedule shipping.



Amazon MWAA (Managed Workflows for Apache Airflow)

Data Pipeline Orchestration

- Managed Apache Airflow service
- Create, schedule, and monitor data pipelines
- Python-based workflow definition (DAGs)
- Integration with AWS data services

Use Case: Schedule daily ETL jobs to process sales data and load it into Redshift.

Data Analytics

Process, analyze, and visualize large-scale data.



Amazon EMR (Elastic MapReduce)

Big Data Processing

- Process vast amounts of data with Spark, Hadoop, Presto, Hive
- Scalable clusters
- Integration with S3, DynamoDB
- Cost-effective with Spot Instances

Use Case: Process terabytes of click-stream data to identify user behavior patterns.



Amazon Athena

Serverless SQL Queries on S3

- Query data directly in S3 using standard SQL
- No infrastructure to manage
- Pay per query (data scanned)
- Integration with Glue Data Catalog

Use Case: Analyze years of server logs stored in S3 without moving the data.



AWS Glue

Serverless ETL

- Extract, Transform, Load (ETL) service
- Data catalog and schema discovery
- Python or Scala-based ETL scripts
- Job scheduling and monitoring

Use Case: Automatically crawl, catalog, and transform data from multiple sources before loading into Redshift.



Amazon Redshift

Petabyte-Scale Data Warehouse

- Columnar storage for fast analytical queries
- Massively parallel processing (MPP)
- Integration with BI tools
- Expensive but powerful for large-scale analytics

Use Case: Analyze petabytes of historical sales data to identify trends and forecasts.



Amazon QuickSight

Business Intelligence & Dashboards

- Serverless BI service
- Interactive dashboards and visualizations
- ML-powered insights
- Embedded analytics

Use Case: Create executive dashboards showing real-time sales metrics and trends.



Amazon Kinesis

Real-Time Data Streaming

- **Kinesis Data Streams:** Capture and store streaming data
- **Kinesis Firehose:** Load streaming data into S3, Redshift, Elasticsearch
- **Kinesis Data Analytics:** Analyze streaming data with SQL
- Real-time processing of logs, IoT data, clickstreams

Use Case: Process millions of website clickstream events per second in real-time to update live dashboards.

Monitoring & Management

Monitor, log, and manage your AWS infrastructure.

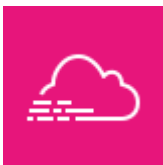


Amazon CloudWatch

Monitoring & Observability

- Collect and track metrics
- Log aggregation and search
- Alarms and notifications
- Dashboards for visualization
- Application Insights

Use Case: Set an alarm to notify you when EC2 CPU usage exceeds 80% for 5 minutes.



AWS CloudTrail

Audit & Compliance Logging

- Track user activity and API usage
- Who did what, when, from where
- Governance, compliance, risk auditing
- Integration with CloudWatch Logs

Use Case: Investigate who deleted a critical S3 bucket and when it happened.



AWS Config

Resource Configuration Tracking

- Record and evaluate AWS resource configurations
- Compliance checking against rules
- Configuration history and change tracking
- Automated remediation

Use Case: Ensure all S3 buckets are private and automatically flag non-compliant resources.



AWS X-Ray

Distributed Application Tracing

- Trace requests across microservices
- Identify performance bottlenecks
- Service map visualization
- Error and exception analysis

Use Case: Debug why some API requests take 10 seconds while others take 1 second by tracing the request flow.

CI/CD & Developer Tools

Automate your software delivery process.



AWS CodeBuild

Continuous Integration

- Compile source code
- Run tests
- Produce deployable artifacts
- Fully managed build service
- Integration with GitHub, CodeCommit, Bitbucket

Use Case: Automatically build and test your code every time you push to the main branch.



AWS CodeDeploy

Automated Deployment

- Deploy to EC2, Lambda, ECS, on-premises servers
- Blue/green and rolling deployments
- Automatic rollback on failure
- Deployment health tracking

Use Case: Deploy new application versions to production with zero downtime using blue/green deployments.



AWS CodePipeline

Continuous Delivery

- Orchestrate build, test, and deploy phases
- Visual pipeline designer
- Integration with third-party tools (GitHub Actions, Jenkins)
- Parallel and sequential actions

Use Case: Create a pipeline that automatically builds, tests, and deploys your code from GitHub to production.



AWS CloudFormation

Infrastructure as Code (IaC)

- Define AWS resources using JSON or YAML templates
- Version control your infrastructure
- Stack management (create, update, delete)
- Drift detection
- Change sets for preview

Use Case: Define your entire application infrastructure (VPC, EC2, RDS, S3) in one template and deploy it consistently across environments.



AWS CDK (Cloud Development Kit)

Programmatic Infrastructure as Code

- Define infrastructure using TypeScript, Python, Java, C#, Go
- Compiles to CloudFormation templates
- Reusable constructs and patterns
- Type safety and IDE autocomplete

Use Case: Use Python to define your infrastructure with loops, conditionals, and object-oriented patterns.



AWS Amplify

Full-Stack App Development Platform

- Frontend framework (React, Vue, Angular, Next.js)
- Backend features: Authentication, APIs, Storage
- Hosting with CI/CD
- Admin UI for non-developers

Use Case: Build and deploy a full-stack React application with authentication and API in minutes.



AWS AppSync

Managed GraphQL Service

- Create flexible GraphQL APIs
- Real-time data synchronization
- Offline data access
- Connect to DynamoDB, Lambda, RDS, HTTP APIs

Use Case: Build a real-time chat application where messages appear instantly across all connected clients.

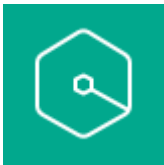


Amazon WorkSpaces

Virtual Desktops (DaaS)

- Managed, secure Desktop-as-a-Service
- Windows and Linux desktops
- Stream applications to any device
- Pay-as-you-go pricing

Use Case: Provide remote developers with secure, pre-configured development environments.



Amazon Q

AI Coding Assistant

- IDE integration (VS Code, IntelliJ)
- Code suggestions and completions
- Security scanning
- AWS service recommendations

Use Case: Get inline code suggestions while building AWS Lambda functions.



AWS MCP (Model Context Protocol)

LLM Context Integration

- Enable LLMs to query AWS resources safely
- Provides context about your AWS environment
- Security and permission controls

Use Case: Ask an AI assistant questions about your AWS infrastructure and get accurate, context-aware responses.

Identity & Networking

Secure access and network isolation for your AWS resources.



IAM (Identity and Access Management)

Access Control

- Users, Groups, Roles, Policies
- Fine-grained permissions
- Multi-factor authentication (MFA)
- Temporary security credentials
- Service control policies (SCPs)

Use Case: Create a developer role that has read-only access to production resources but full access to development.



AWS IAM Identity Center

Workforce Single Sign-On

- Formerly AWS SSO
- Centralized access to multiple AWS accounts
- Integration with Active Directory, Okta, etc.
- Application access management

Use Case: Allow employees to log in once and access all AWS accounts and business applications.



Amazon VPC (Virtual Private Cloud)

Isolated Cloud Network

- Logically isolated network in AWS
- Complete control over IP ranges, subnets, route tables
- Internet Gateway, NAT Gateway
- Network ACLs and Security Groups

Use Case: Create a private network for your application with a public subnet for web servers and a private subnet for databases.

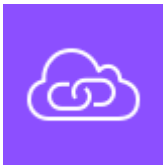


AWS VPN

Secure Network Connection

- **Site-to-Site VPN:** Connect on-premises network to AWS
- **Client VPN:** Remote user access to AWS resources
- IPsec encryption
- High availability with multiple tunnels

Use Case: Securely connect your corporate data center to your AWS VPC.



AWS PrivateLink

Private Connectivity

- Access AWS services without internet exposure
- VPC-to-VPC communication
- Third-party SaaS integration
- No public IPs required

Use Case: Connect to S3 from your private subnet without routing traffic through the internet.

Quick Reference

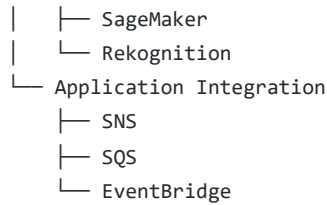
By Use Case

Use Case	Recommended Service
Host a static website	S3 + CloudFront + Route 53
Run a web application	EC2 / ECS / Lambda + ALB
Relational database	RDS (PostgreSQL, MySQL) or Aurora
NoSQL database	DynamoDB or DocumentDB
File storage	S3 (object) or EFS (shared filesystem)
User authentication	Cognito
API creation	API Gateway + Lambda
Serverless compute	Lambda

Container orchestration	ECS with Fargate or EKS
Caching	ElastiCache (volatile) or MemoryDB (durable)
Queue messaging	SQS
Pub/Sub messaging	SNS
Workflow orchestration	Step Functions
Data warehouse	Redshift
Real-time analytics	Kinesis
Machine learning	SageMaker
Infrastructure as Code	CloudFormation or CDK
CI/CD	CodePipeline + CodeBuild + CodeDeploy
Monitoring	CloudWatch
Security audit	CloudTrail

Service Categories at a Glance

AWS Services
├─ Compute
├─ EC2
├─ Lambda
└─ ECS/EKS
├─ Storage & Database
├─ S3
├─ RDS/Aurora
└─ DynamoDB
├─ Networking & CDN
├─ VPC
├─ CloudFront
└─ Route 53
├─ Security & Identity
├─ IAM
├─ Cognito
└─ WAF
├─ Developer Tools
├─ CodePipeline
├─ CloudFormation
└─ X-Ray
├─ Analytics
├─ Athena
├─ Redshift
└─ Kinesis
├─ AI/ML
└─ Bedrock



Learning Path Recommendations

Beginner (Start Here)

1. **IAM** - Secure access management
2. **S3** - Object storage
3. **EC2** - Virtual servers
4. **RDS** - Managed databases
5. **VPC** - Networking basics

Intermediate

6. **Lambda** - Serverless compute
7. **API Gateway** - API management
8. **DynamoDB** - NoSQL database
9. **CloudFront** - CDN
10. **CloudFormation** - Infrastructure as Code

Advanced

11. **ECS/EKS** - Container orchestration
12. **Step Functions** - Workflow orchestration
13. **Kinesis** - Real-time streaming
14. **SageMaker** - Machine learning
15. **X-Ray** - Distributed tracing

Conclusion

These **50+ AWS services** represent the core knowledge you need to build, deploy, and scale modern cloud applications. While AWS offers over 300 services, mastering these fundamentals will make you an effective AWS practitioner.

 **Pro Tip:** Start with the basics (IAM, EC2, S3, RDS), then expand to serverless (Lambda, DynamoDB, API Gateway) as your needs grow.

Additional Resources

-  [Official AWS Documentation](#)
-  [AWS Training & Certification](#)
-  [AWS Well-Architected Framework](#)
-  [AWS Pricing Calculator](#)
-  [AWS Free Tier](#)

Last Updated: January 2026

Version: 1.0